Dashboard Mode

- Real Data — VitalDB
- Synthetic — Deep Dive

Patient Simulator

View Mode

- Single Patient
- Side-by-Side
- CA vs No-CA Overlay

Noise level

1.00

Patient A

Seed A

42

Duration A (min)

480

CA Event A

CA Onset A (min)

380

HR Rise Rate A

0.12

BP Drop Rate A

0.15

SpO2 Drop Rate A

0.06

INTRAOPERATIVE CARDIAC ARREST RISK MONITOR

SYNTHETIC DEEP DIVE

Configurable Patient Simulator · All waveforms algorithmically generated | 12 Apr 2026 · 14:15 | Side-by-Side

Sachu Mon P. Sajeev · MSc Data Science & AI · TSI University Riga · 2026

TAN · AUROC	LightGBM · AUROC	TAN · AUPRC
0.9937	0.9073	0.6027
95% CI [0.9909–0.9961]	60-min window	240-min window

Synthetic Mode — All waveforms are algorithmically generated from configurable parameters. Adjust deterioration rates, onset timing, and noise in the sidebar. Perfect for thesis presentation and clinical model explanation.

RISK COMPARISON SUMMARY

Patient A — CA Event

52.4%

HIGH

Patient B — No CA

4.1%

LOW

Patient #42

CA EVENT · SYNTHETIC · Onset: 380min

CA RISK · 60MIN WINDOW

52.4%

HIGH

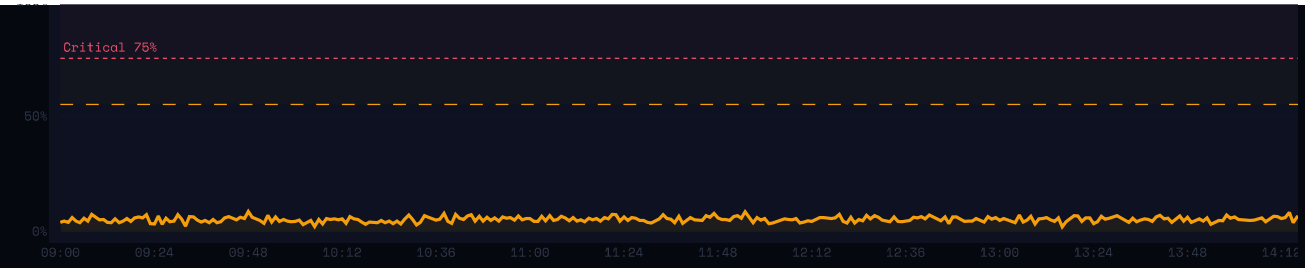
HEART RATE	SPO₂	ETCO₂	MEAN BP	SYSTOLIC BP	DIASTOLIC BP
86	95	34	69	110	49
bpm	%	mmHg	mmHg	mmHg	mmHg
→ OK	↓ LOW	↓ LOW	↓ LOW	→ OK	↓ LOW
→0.2/30m	→0.8/30m	↘2.0/30m	↘3.3/30m	↘3.0/30m	↘2.9/30m
Range: 69–86	Range: 94–98	Range: 33–40	Range: 68–90	Range: 109–130	Range: 48–71
Normal: 50–100	Normal: 95–100	Normal: 35–45	Normal: 70–105	Normal: 90–140	Normal: 60–90

DETERIORATION PHASE TIMELINE

STABLE PHASE	EARLY DETERIORATION	ACTIVE CRISIS
380min	40min	41min
0–380min mark	380–420min mark	420–461min mark

RISK SCORE OVER TIME

100%



VITAL SIGNAL TRENDS

Waveform algorithmically generated from configurable parameters.

Heart Rate (bpm) · normal 50–100

SpO₂ (%) · normal 95–100

EtCO₂ (mmHg) · normal 35–45

Mean BP (mmHg) · normal 70–105

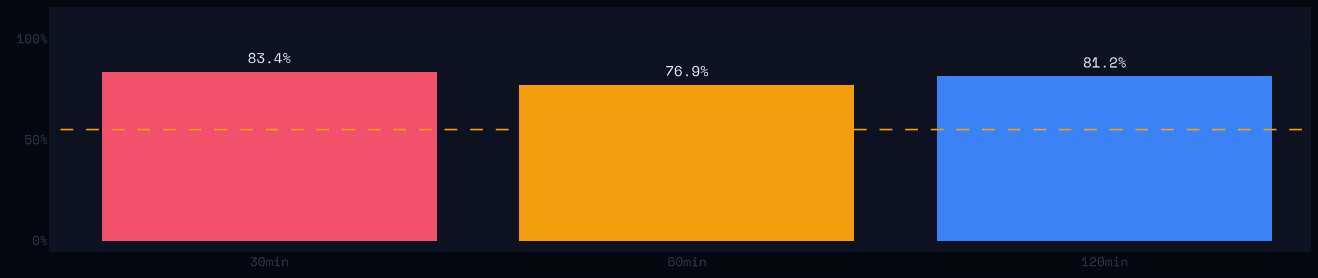
Systolic BP (mmHg) · normal 90–140

Diastolic BP (mmHg) · normal 60–90

CHANGE POINT LOG

No significant change points detected.

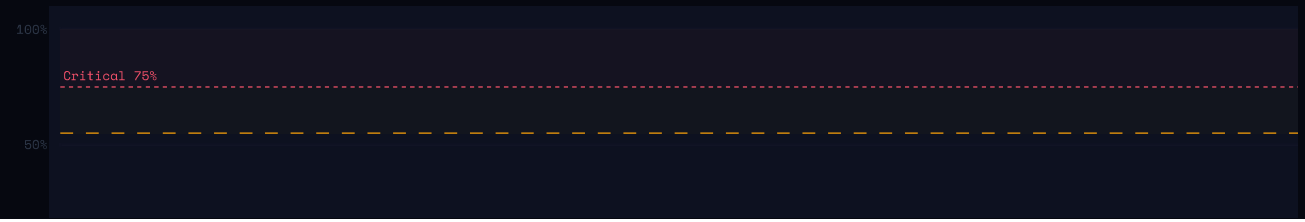
RISK ACROSS AVAILABLE WINDOWS



Patient #77 ✓ NO CA SYNTHETIC

CA RISK · 60MIN WINDOW	HEART RATE	SPO₂	ETCO₂	MEAN BP	SYSTOLIC BP	DIASTOLIC BP
4.1 %	71	98	38	85	127	66
LOW	bpm	%	mmHg	mmHg	mmHg	mmHg
	→ OK	→ OK	→ OK	→ OK	→ OK	→ OK
	→0.1/30m	→0.0/30m	→0.1/30m	→0.7/30m	→0.3/30m	→0.2/30m
	Range: 66–78	Range: 97–98	Range: 36–39	Range: 82–88	Range: 120–131	Range: 61–71
	Normal: 50–100	Normal: 95–100	Normal: 35–45	Normal: 70–105	Normal: 90–140	Normal: 60–90

RISK SCORE OVER TIME



VITAL SIGNAL TRENDS

🚀 Waveform algorithmically generated from configurable parameters.

Heart Rate (bpm) · normal 50–100

SpO_2 (%) · normal 95–100

$EtCO_2$ (mmHg) · normal 35–45

Mean BP (mmHg) · normal 70–105

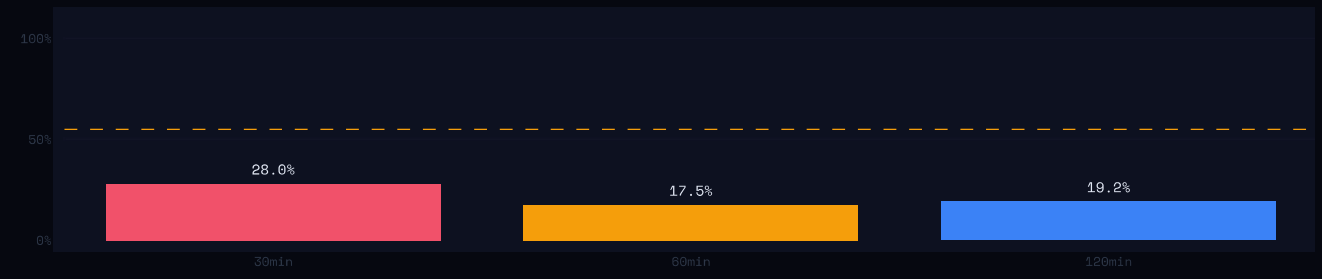
Systolic BP (mmHg) · normal 90–140

Diastolic BP (mmHg) · normal 60–90

CHANGE POINT LOG

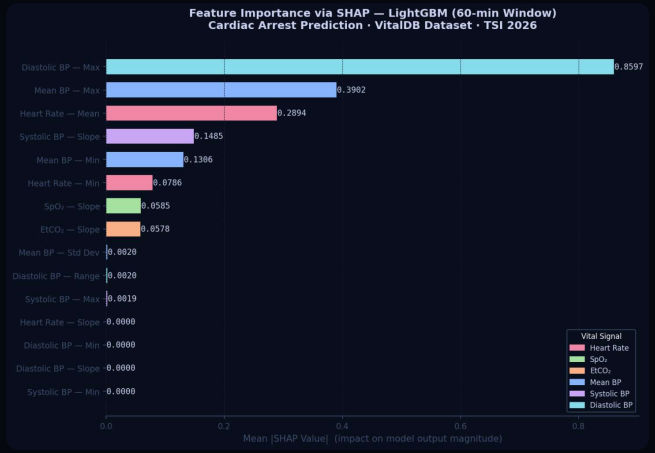
No significant change points detected.

RISK ACROSS AVAILABLE WINDOWS

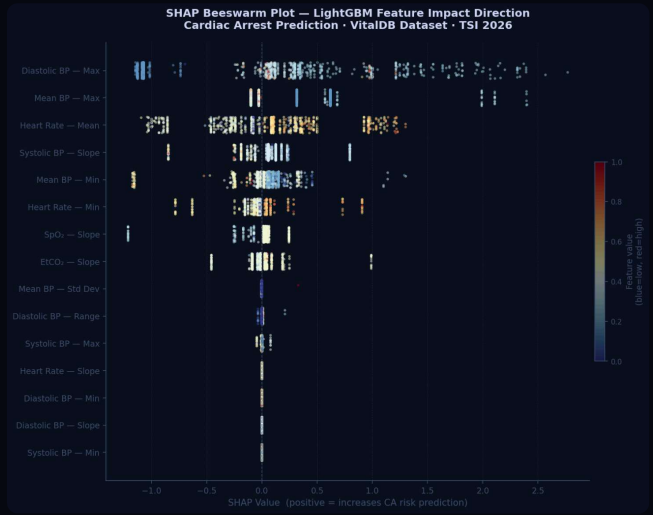


🔬 SHAP Feature Interpretability — LightGBM (60-min Window)

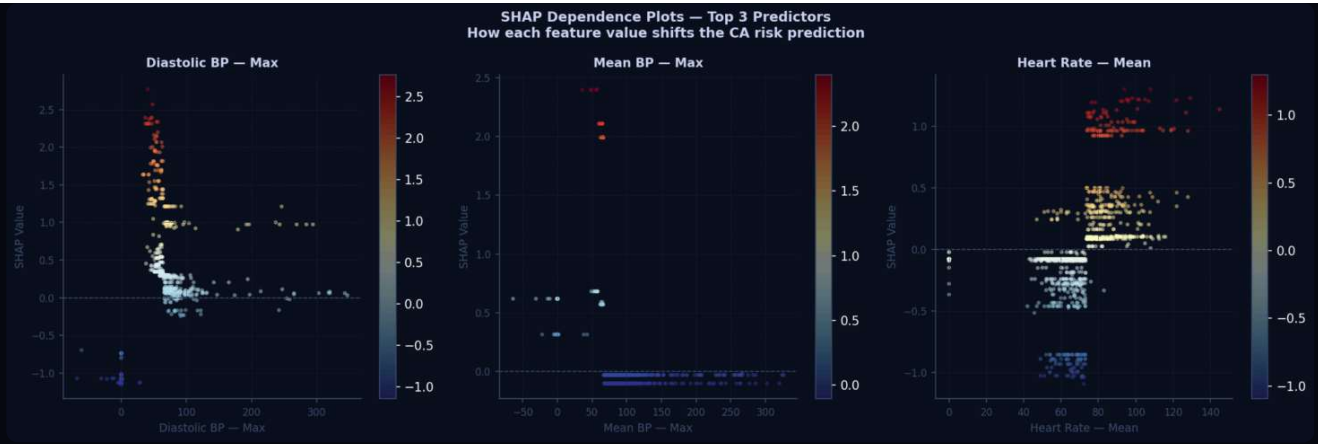
SHapley Additive exPlanations · TreeExplainer · AUROC 0.9073 · VitalDB Dataset · TSI 2026



Mean |SHAP| — Top 15 Features



SHAP Beeswarm — Feature Impact Direction



SHAP Dependence — Top 3 predictors

🔬 **How to read these charts:**

Bar chart — average magnitude of each feature's impact. Longer bar = stronger predictor.

Beeswarm — each dot is one patient. Red dots pushed right = high value increases CA risk. Blue pushed left = decreases risk.

Key finding — Diastolic BP Max, Mean BP Max, and Heart Rate Mean are the top 3 predictors. Run `python shap_analysis.py` to regenerate.

🟢 **240min** — Full 4-hour window | 🔵 **120min** — 2-hour window | 🟡 **60min** — 1-hour window | 🔴 **30min** — Minimal 30-min window

🔴 CRITICAL ≥75% — Alert physician immediately | 🟠 HIGH 50–75% — Increase monitoring | 🟡 MODERATE 25–50% — Watch closely | 🔵 LOW <25% — Routine monitoring